

EXPERIMENT 19. Display Dimensions/Shape and Column Names of World Alcohol Consumption Dataset

Aim:

To write a Pandas program to display the **dimensions or shape** of the World Alcohol Consumption dataset and extract its **column names**.

Program:

CO4 > exp19.py > ...

```
1 import pandas as pd
2 data = {
3     'Year': [2010, 2011, 2012, 2013, 2014],
4     'WHO region': ['Europe', 'Americas', 'Asia', 'Africa', 'Europe'],
5     'Country': ['France', 'USA', 'India', 'Nigeria', 'Germany'],
6     'Beverage Types': ['Wine', 'Beer', 'Spirits', 'Beer', 'Wine'],
7     'Display Value': [12.5, 10.2, 4.5, 3.8, 11.7]
8 }
9 df = pd.DataFrame(data)
10 print("World Alcohol Consumption Dataset:")
11 print(df)
12 print("\nShape of the dataset:")
13 print(df.shape)
14 print("\nColumn names:")
15 print(df.columns.tolist())
```

Output

World Alcohol Consumption Dataset:

	Year	WHO region	Country	Beverage Types	Display Value
0	2010	Europe	France	Wine	12.5
1	2011	Americas	USA	Beer	10.2
2	2012	Asia	India	Spirits	4.5
3	2013	Africa	Nigeria	Beer	3.8
4	2014	Europe	Germany	Wine	11.7

Shape of the dataset:

(5, 5)

Column names:

['Year', 'WHO region', 'Country', 'Beverage Types', 'Display Value']

PS C:\Users\dhars\Documents\Query processing\lab_questions> █

Result

Thus, the **dimensions of the World Alcohol Consumption dataset** and its **column names** were successfully displayed using Pandas.

EXPERIMENT 20. Find the Index of a Given Substring in a DataFrame Column

Aim:

To write a Pandas program to find the **index/position of a given substring** in a DataFrame column.

Program:

```
exp20.py > ...
1  import pandas as pd
2  data = {
3      'name': ['John Smith', 'David Brown', 'Smith Wilson', 'Alice Smith']
4  }
5  df = pd.DataFrame(data)
6  df['index'] = df['name'].str.find('Smith')
7  print(df)
8
```

Output:

```
PS C:\Users\dhars\Documents\Query processing\lab_questions> & C:\Us
on.exe "c:/Users/dhars/Documents/Query processing/lab_questions/exp
      name  index
0   John Smith    5
1  David Brown   -1
2 Smith Wilson    0
3  Alice Smith    6
PS C:\Users\dhars\Documents\Query processing\lab_questions> █
```

Result:

Thus, the index position of the given substring **"Smith"** was successfully found using the Pandas `str.find()` function. A value of **-1** indicates that the substring was not found.